

Subnetting

A subnetwork or subnet is a **logical subdivision** of an IP network. The practice of **dividing a network into two or more networks** is called subnetting.

Without subnetting, you might waste thousands of IP addresses. If you have 50 computers but your network supports 65,000 addresses, you're wasting 99.9% of your addresses.

Network and Hosts in Classes:

	8 Bits (Octet)	8 Bits (Octet)	8 Bits (Octet)	8 Bits (Octet)
Class A	Network	Hosts	Hosts	Hosts
Class B	Network	Network	Hosts	Hosts
Class C	Network	Network	Network	Hosts

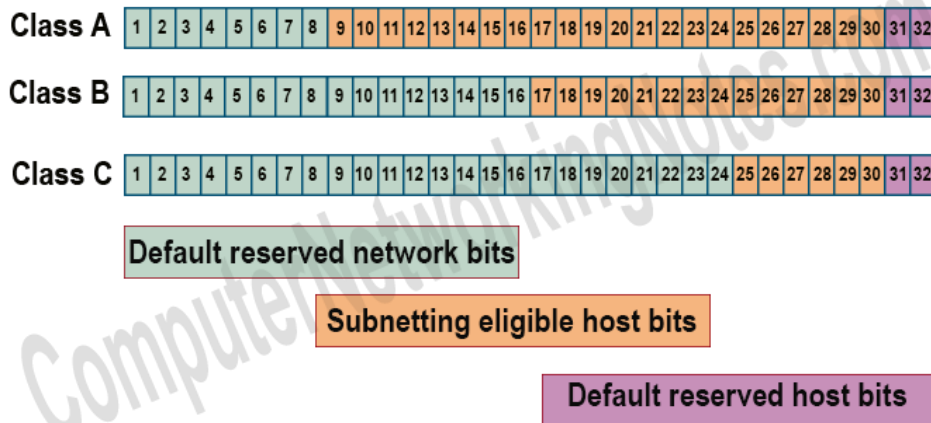
Number of Hosts Combination in Classes:

Lets see the number of possible host combinations in each class.

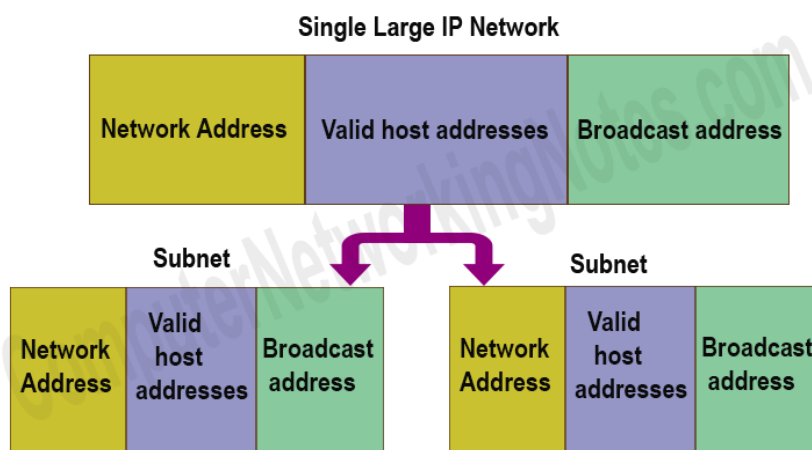
In class C, 2^8 hosts in each network are possible ie., equals to 256 hosts in each network .

	Possible Hosts
Class A	$256 \times 256 \times 256 = 16,777,216$
Class B	$256 \times 256 = 65,536$
Class C	256

Eligible Bits for Subnetting:



Network and Broadcast Addresses:



CIDR Notation

Instead of writing 255.255.255.0, we write /24 The

/24 means "24 bits are used for the network"

Common Subnet Masks:

- /24 = 255.255.255.0 = 254 usable hosts
- /25 = 255.255.255.128 = 126 usable hosts
- /26 = 255.255.255.192 = 62 usable hosts • /27 = 255.255.255.224 = 30 usable hosts
- /28 = 255.255.255.240 = 14 usable hosts
- /29 = 255.255.255.248 = 6 usable hosts
- /30 = 255.255.255.252 = 2 usable hosts (used for point-to-point links)

Subnet Masks

Subnet mask	Binary	Networks	Hosts
255.255.255.0	11111111.11111111.11111111.00000000	1	254
255.255.255.128	11111111.11111111.11111111.10000000	2	126
255.255.255.192	11111111.11111111.11111111.11000000	4	62
255.255.255.224	11111111.11111111.11111111.11100000	8	30
255.255.255.240	11111111.11111111.11111111.11110000	16	14
255.255.255.248	11111111.11111111.11111111.11111000	32	6
255.255.255.252	11111111.11111111.11111111.11111100	64	2
255.255.255.254	11111111.11111111.11111111.11111110	128	0

Example

You have the network 192.168.1.0/24 and need to create 4 subnets.

Step 1: Determine How Many Bits to Borrow

Formula: $2^n \geq \text{number of subnets needed}$

We need 4 subnets:

- $2^1 = 2$ (not enough)
- $2^2 = 4$ (perfect!)

So we borrow **2 bits** from the host portion.

Step 2: Calculate New Subnet Mask

Original: /24 (255.255.255.0) Borrowed bits: 2 New: /26 (255.255.255.192)

Step 3: Calculate Subnet Block Size

Formula: $256 - \text{last octet of subnet mask}$

$256 - 192 = 64$

This means each subnet has 64 addresses.

Step 4: List Your Subnets

Starting from 0, add the block size:

Subnet	Network Address	First Usable	Last Usable	Broadcast	Hosts
--------	-----------------	--------------	-------------	-----------	-------

1	192.168.1.0/26	192.168.1.1	192.168.1.62	192.168.1.63	62
2	192.168.1.64/26	192.168.1.65	192.168.1.126	192.168.1.127	62
3	192.168.1.128/26	192.168.1.129	192.168.1.190	192.168.1.191	62
4	192.168.1.192/26	192.168.1.193	192.168.1.254	192.168.1.255	62

- **Network Address:** First address (can't be assigned to devices)
- **Broadcast Address:** Last address (can't be assigned to devices)
- **Usable Hosts:** Total addresses - 2 (network and broadcast)

Example 1:

Let's suppose, we have

IP address: 100.10.20.0

Number of required hosts: 100 hosts per subnet
Number of required subnets: 30 subnets

Perform subnetting and tell which subnet contains the given IP address.

Step 1:

Identify Class + Upper Bounds

- This is class **A**. (IP: 100.10.20.0 → 1st Octet = 100 = **CLASS A**)
- **Hosts Needed:** 100 → Upper Bound = $2^7 = 128$ ($128 - 2 = 126$ usable > 100)
- **Subnets Needed:** 30 → Upper Bound = $2^5 = 32$

Step 2:

Write a subnet of the class of the ip address.

255.0.0.0

11111111.00000000.00000000.00000000

Step 3:

Check if subnetting is possible.

Total Required: $2^7 \times 2^5 = 4096$

Class A Capacity: $2^{(24)} = 16,777,216$ hosts

$4096 \ll 16M \rightarrow$ SUBNETTING POSSIBLE

Step 4:

Borrow bits from the left of the subnet mask from host octets.

11111111.00000000.00000000.00000000



Now we have an upper bound 2^5 of subnets so we will borrow 5 bits from hosts for subnets.

11111111.00000000.00000000.00000000
11111111.11110000.00000000.00000000

New Subnet Mask: **255.248.0.0**

Step 5:

Calculate Subnet Block Size

Formula: 256 - last octet of subnet mask
 $256 - 248 = 8$

This means each we will have a range of 8.

Step 6:

Creating the subnets.

Tip: Start creating subnets from the octet you borrowed zeros. In our case it's the second octet and class A.

First subnet:

100.0.0.0 - 100.7.255.255

The range is 8 so including 0 till 7 (0,1,2,3,4,5,6,7) makes 8 numbers.

Network Address: 100.0.0.0

Broadcast Address: 100.7.255.255

Second subnet:

100.8.0.0 - 100.15.255.255

Network Address: 100.8.0.0

Broadcast Address: 100.15.255.255

Third subnet:

100.16.0.0 - 100.23.255.255

Network Address: 100.16.0.0

Broadcast Address: 100.23.255.255

Fourth subnet:

100.24.0.0 - 100.31.255.255

Network Address: 100.24.0.0

Broadcast Address: 100.31.255.255

Fifth subnet:

100.32.0.0 - 100.39.255.255

Network Address: 100.32.0.0

Broadcast Address: 100.39.255.255

So on.....

Assign the subnets to topology along with a new subnet mask **255.248.0.0**

Below we only show first 10 subnets:

Subnet #	Network Address	First Usable	Last Usable	Broadcast	Hosts
1	100.0.0.0/13	100.0.0.1	100.7.255.254	100.7.255.255	524288
2	100.8.0.0/13	100.8.0.1	100.15.255.254	100.15.255.255	524288
3	100.16.0.0/13	100.16.0.1	100.23.255.254	100.23.255.255	524288
4	100.24.0.0/13	100.24.0.1	100.31.255.254	100.31.255.255	524288
5	100.32.0.0/13	100.32.0.1	100.39.255.254	100.39.255.255	524288
6	100.40.0.0/13	100.40.0.1	100.47.255.254	100.47.255.255	524288
7	100.48.0.0/13	100.48.0.1	100.55.255.254	100.55.255.255	524288
8	100.56.0.0/13	100.56.0.1	100.63.255.254	100.63.255.255	524288
9	100.64.0.0/13	100.64.0.1	100.71.255.254	100.71.255.255	524288
10	100.72.0.0/13	100.72.0.1	100.79.255.254	100.79.255.255	524288

Step 7:

Which Subnet Contains 100.10.20.0...

Practice Examples: solve on papers

1. IP address: 172.168.1.0
Number of required hosts: 160 hosts per subnet
Number of required subnets: 60 subnets
Perform subnetting and tell which subnet contains the given IP address.

2. IP address: 10.10.20.0
Number of required hosts: 100 hosts per subnet
Number of required subnets: 100 subnets
Perform subnetting and tell which subnet contains the given IP address.

3. IP address: 192.168.29.1
Number of required hosts: 100 hosts per subnet
Number of required subnets: 60 subnets
Perform subnetting and tell which subnet contains the given IP address.

4. IP address: 220.162.49.1
Number of required hosts: 30 hosts per subnet
Number of required subnets: 60 subnets
Perform subnetting and tell which subnet contains the given IP address.